Yatra Booking Agent for Vaishno Devi – TOURISM_GPT

(A Unified Digital Platform for Pilgrimage Trip Planning and Booking)

Abstract

Anyone who has tried to plan a Vaishno Devi trip knows the drill: one tab open for the Shrine Board's registration site, another for flight or train bookings, a third for a hotel in Katra, and usually a phone call or two to confirm something none of those sites actually tell you. There is no single place that pulls all of this together, and that gap costs pilgrims time and, more often than not, leads to choices made with incomplete information rather than a real comparison of what's available.

This project sets out to close that gap. The Yatra Booking Agent lets a user enter their trip details once, where they're travelling from, how many people, which mode of transport they'd prefer, the type of darshan, and the date — and get back a ranked set of options that combine transport and darshan slots together, rather than as two separate searches. From there, the same flow carries the user through hotel selection, sightseeing suggestions for the area around Katra, and a mock payment step, so the whole trip is planned in one place instead of four.

The application is built in Python using Stream lit, and it draws on a mix of real and simulated data depending on what's actually available publicly. Darshan pricing, transport connectivity, and weather come from real, verified sources; things like live seat counts or hotel room inventory, which no public system exposes, are simulated but clearly marked as such. The result is a working prototype that shows what a properly unified yatra planning tool could look like, built on data that's honest about its own limits.

Problem Statement:

Right now, a pilgrim planning a trip has to check the Shrine Board portal for darshan availability, a transport site for flights or trains, and a separate hotel platform — with nothing tying these together. That fragmentation isn't just inconvenient. It means trip decisions get made without full visibility into cost or timing across the whole journey, and there's no single record of what was planned versus what actually got booked. It's a gap in the experience, and also an opportunity: nobody currently owns this end-to-end journey, which leaves room for a consolidated tool to fill it.

Objectives and Scope:

The current phase of this project covers multi-modal transport search across flight, train, and bus, matched against darshan slot availability; hotel selection in Katra; sightseeing recommendations for the surrounding region; and supporting information like weather and local taxi guidance, all wrapped in a mock payment flow so the experience feels complete end to end.

What it doesn't cover yet, deliberately: live integration with the Shrine Board's actual registration system, real airline or IRCTC booking, real payment processing, or production-grade user authentication. Those are the next phase, not this one.

Deliverables:

- A working booking agent that searches and ranks flight, train, and bus options against available darshan slots, based on what the user actually asked for.
- One consolidated planning view, in place of separately checking transport sites, the Shrine Board portal, and hotel platforms.
- A full booking flow — transport, darshan slot, hotel, sightseeing — that takes a user from initial search through to a confirmed (mock) booking.
- Practical planning support, including live and seasonal weather forecasts and researched local taxi fares, so decisions are grounded in real conditions rather than guesswork.
- A data layer that's honest about its own sourcing — real figures where they exist, clearly labelled simulation where they don't — which gives future work a clean foundation to build on rather than a black box to untangle.

Future Scope:

There's a reasonably clear path beyond Vaishno Devi itself — the same approach could extend to other pilgrimage sites with similar planning challenges. Multilingual support would matter a great deal here given who the actual user base is, and integrating UPI alongside card payments would make a live version far more usable for most Indian travellers than a card-only checkout.

Keywords: Pilgrimage Trip Planning, Multi-Modal Travel Booking, Stream lit Application, Tourism Technology, Data-Driven Travel Recommendation